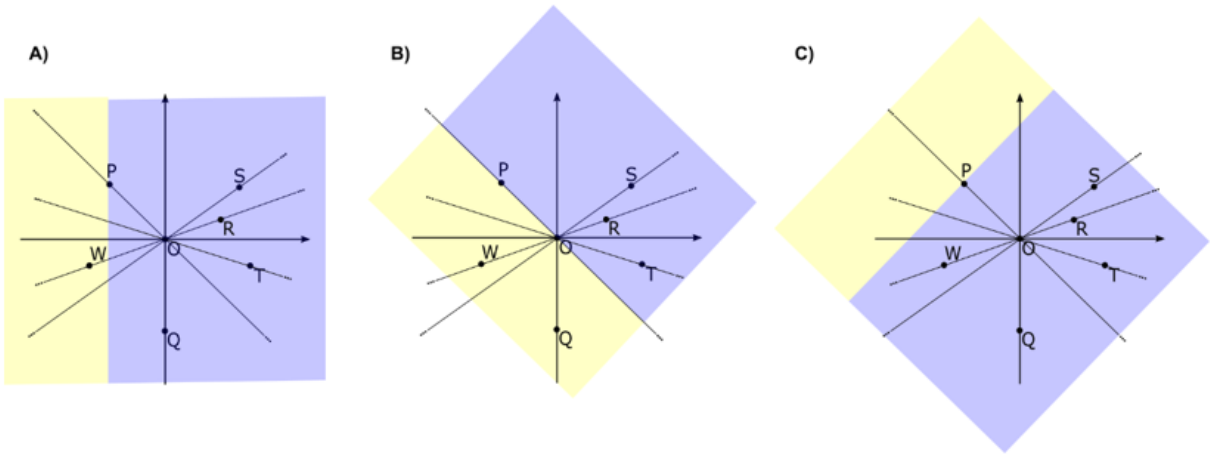


Figure 8.8 Points in a cartesian plane. Lines through the origin and each point are shown. A few possible partitionings into left and right of P are shown.
(A) Splitting the plane with a vertical line passing through P.
(B) Drawing a line passing through P and the origin, will define two semi-spaces on the two sides of the line.
(C) Using the same line as B but partitioning points whose projection on that line is left or right of P.

This solution can look fine while using P as a pivot, but if we take other points, a couple of drawbacks will emerge:

- Looking at point R, in figure 8.9, if we draw a vertical line, parallel to the y axis, and use the x coordinate to partition points, we get W, P, O, Q, and U in the left partition, and S and T in the right one. This is, despite U and S being much closer than U and O or S and T, they end up in different partitions (while the other 2 pair of points are in the same partition).
- If we consider O, which partition will Q be? And what about any other point on the y axis? We can arbitrarily